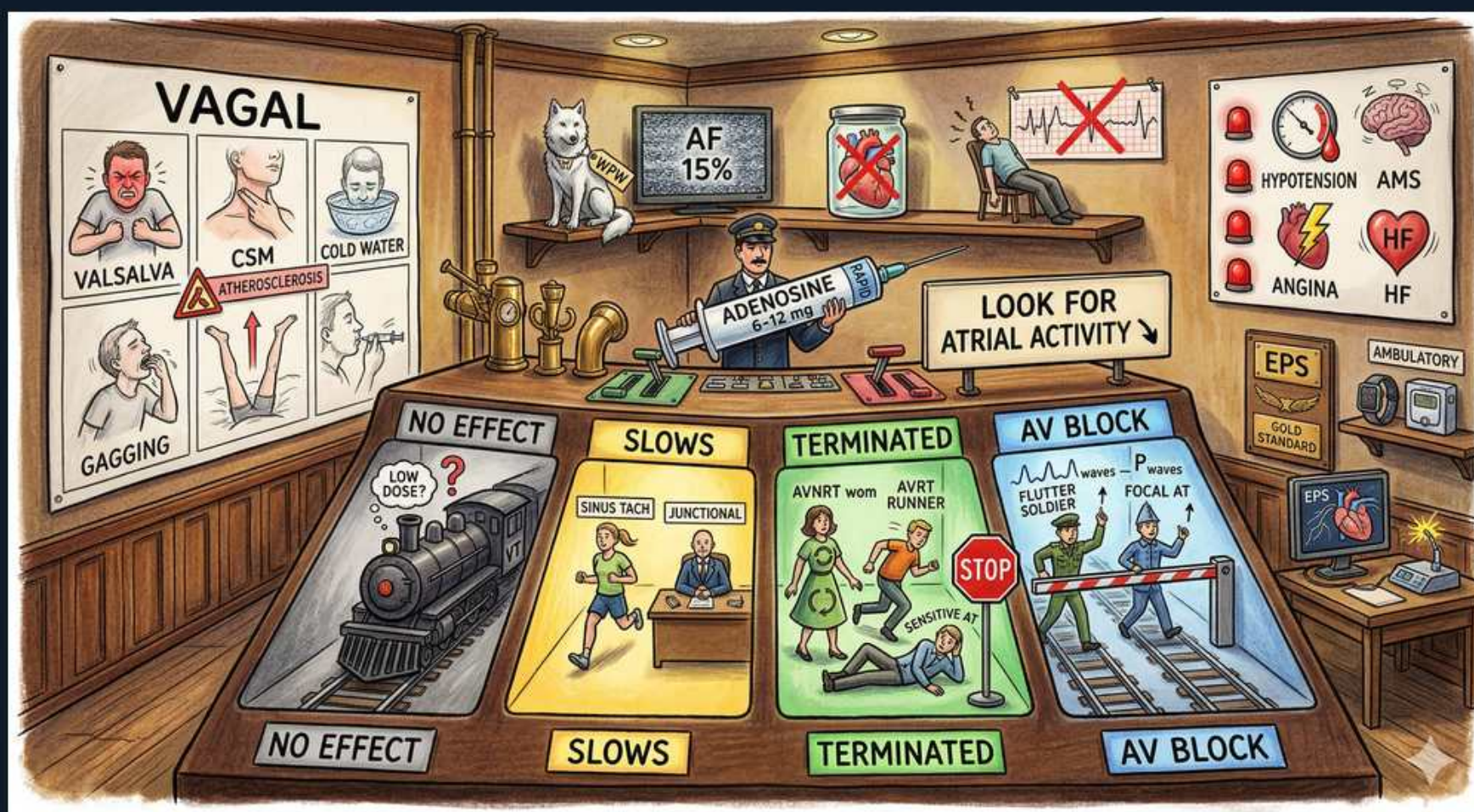


Arrhythmias — Vagal Maneuvers & Adenosine in SVT



1. Vagal maneuvers

WHAT YOU SEE

A large 'VAGAL' poster top-left illustrating the maneuvers — the poster = the first-line vagal toolkit (Valsalva, carotid massage, etc.) that boosts vagal tone to break re-entry.

WHAT IT ENCODES

Vagal maneuvers are FIRST-LINE for a hemodynamically STABLE, regular, narrow-complex tachycardia (suspected AVNRT/AVRT). They increase parasympathetic tone, which slows AV-nodal conduction and can break the re-entry circuit, terminating the arrhythmia. They are also diagnostic: the response (terminate vs slow vs unmask atrial activity) tells you the mechanism.

BOARD TRAP

Vagal maneuvers (and adenosine) are the WRONG first step if the patient is UNSTABLE — unstable narrow-complex tachycardia goes straight to synchronized cardioversion. They also won't terminate AF, flutter, or AT (AV-node-independent rhythms).

2. Valsalva

WHAT YOU SEE

A straining, red-faced man bearing down, labelled 'VALSALVA' — the strain against a closed glottis = the Valsalva maneuver that reflexively spikes vagal tone to slow the AV node.

WHAT IT ENCODES

The Valsalva maneuver (forced expiration against a closed glottis) raises intrathoracic pressure and reflexively boosts vagal tone to slow AV conduction. The MODIFIED Valsalva — strain semi-recumbent, then immediately lay the patient flat and passively raise the legs to 45 degrees — significantly improves conversion rates over standard Valsalva and is the preferred technique.

BOARD TRAP

Standard supine Valsalva alone has a fairly low success rate; the modified (leg-lift) Valsalva is the higher-yield, evidence-based version — citing plain Valsalva as equally effective is a trap.

3. Carotid sinus massage

WHAT YOU SEE

A hand pressing on a neck just below the jaw angle, labelled 'CSM' — rubbing the carotid sinus = baroreceptor stimulation that floods the AV node with vagal tone.

WHAT IT ENCODES

Carotid sinus massage stimulates baroreceptors that augment vagal output to the AV node — both diagnostic and therapeutic for AV-node-dependent SVT. Apply firm pressure to ONE carotid sinus (just below the angle of the jaw) for ~5-10 seconds, never both sides simultaneously.

BOARD TRAP

Never massage both carotids at once. CSM is contraindicated with a carotid bruit, known carotid stenosis, or recent (<3 month) TIA/stroke — risk of plaque embolization and stroke.

4. Atherosclerosis caution

WHAT YOU SEE

An 'ATHEROSCLEROSIS' warning tucked under the CSM panel — the plaque warning = why you must not massage a carotid with a bruit/stenosis (risk of dislodging plaque and stroke).

WHAT IT ENCODES

Avoid carotid sinus massage in patients with atherosclerotic disease — a carotid bruit, known stenosis, or prior cerebrovascular event — because mechanical pressure can dislodge plaque and cause an embolic stroke. Auscultate for a bruit before attempting CSM, especially in older patients.

BOARD TRAP

Performing CSM on an elderly patient with a carotid bruit is a board-favorite contraindication violation — choose Valsalva or adenosine instead.

5. Adenosine syringe

WHAT YOU SEE

An officer wielding a big 'ADENOSINE' syringe — the rapid-push syringe = the 6-then-12 mg fast IV bolus that causes a brief total AV block to terminate or unmask the rhythm.

WHAT IT ENCODES

If vagal maneuvers fail, give adenosine: 6 mg rapid IV push followed immediately by a 20 mL saline flush, escalating to 12 mg if no response (a second 12 mg may be repeated). Use a large proximal vein and push fast — its half-life is only ~10 seconds. Adenosine causes transient complete AV block, so it both terminates AV-node-dependent re-entry (AVNRT/AVRT) and serves as a diagnostic by unmasking atrial activity. Warn the patient about a brief, distressing sense of impending doom/flushing/chest pressure.

BOARD TRAP

Slow IV administration fails because adenosine is metabolized within seconds — give it as a rapid push with an immediate flush. Reduce the starting dose (e.g., 3 mg) in patients on dipyridamole or carbamazepine (potentiate it) or with a transplanted heart (denervation hypersensitivity); use higher/standard dosing if on theophylline/caffeine (antagonists).

6. Asthma caution

WHAT YOU SEE

A crossed-out inhaler beside the adenosine — the banned inhaler = adenosine's risk of bronchospasm, so avoid it in severe asthma (use verapamil/diltiazem instead).

WHAT IT ENCODES

Adenosine can provoke bronchospasm via A2B/A3 receptor activation, so use caution or avoid it in patients with reactive airways — severe asthma or active bronchospasm. A non-dihydropyridine calcium-channel blocker (verapamil or diltiazem) or a vagal maneuver is the safer alternative in significant bronchospastic disease.

BOARD TRAP

Reaching for adenosine in a severe/active asthmatic is the trap — bronchospasm can be precipitated. Verapamil/diltiazem is the preferred AV-nodal blocker here (in a stable patient with no WPW).

7. AVOID in WPW-AF

WHAT YOU SEE

A crossed-out delta-wave/AF ECG strip — the forbidden pre-excited-AF tracing = the deadly scenario where AV-nodal blockers shunt conduction down the accessory pathway and can trigger VF.

WHAT IT ENCODES

In pre-excited atrial fibrillation (WPW + AF — irregular, rapid, wide/bizarre QRS), AV-nodal blockers including ADENOSINE, verapamil, diltiazem, beta-blockers, and digoxin are CONTRAINDICATED: blocking the AV node shunts conduction down the accessory pathway, accelerating the ventricular rate and risking degeneration to ventricular fibrillation. Use IV PROCAINAMIDE or ibutilide if stable, or synchronized cardioversion if unstable.

BOARD TRAP

Giving adenosine (or any AV-nodal blocker) in pre-excited AF is dangerous, not therapeutic — it can trigger VF. The tell is an IRREGULAR, wide-complex, very fast tachycardia with varying QRS morphology.

8. No effect → VT

WHAT YOU SEE

A 'NO EFFECT' panel with a heavy train that won't budge for the low dose — no response to adenosine = the rhythm isn't AV-node-dependent, so reconsider VT.

WHAT IT ENCODES

If there is NO RESPONSE to adequate adenosine, the rhythm is likely NOT AV-node-dependent — reconsider VT (ventricular tachycardia is unaffected by AV-nodal blockade), or an AV-node-independent SVT. A non-responding wide-complex tachycardia should be treated as VT.

BOARD TRAP

Interpreting 'no effect from adenosine' as 'the drug failed, give more AV-nodal blockers' is wrong — it is a diagnostic clue pointing toward VT (or atrial flutter/AT). Adenosine can rarely terminate idiopathic RVOT/fascicular VT, but failure to respond should raise VT suspicion, not exclude it.

9. Slows: sinus/junctional

WHAT YOU SEE

A 'SLOWS' panel with sinus-tach and junctional figures briefly easing — only transient slowing then resuming = an AV-node-independent rhythm (sinus/junctional), not a re-entry loop to terminate.

WHAT IT ENCODES

If adenosine only transiently SLOWS the rate and the tachycardia then resumes, the mechanism is AV-node-INDEPENDENT — sinus tachycardia or junctional/automatic rhythm (and sometimes focal AT). The rate is set by the focus, not by an AV-nodal re-entry loop, so blocking the node cannot terminate it.

BOARD TRAP

Transient slowing followed by resumption means the AV node was not the engine — do not keep re-dosing adenosine expecting termination; instead identify and treat the underlying driver (e.g., the cause of sinus tach).

10. Terminated: AVNRT/AVRT

WHAT YOU SEE

A 'TERMINATED' panel with AVNRT/AVRT runners hitting a 'STOP' — abrupt termination = proof of an AV-node-dependent re-entry circuit (AVNRT/orthodromic AVRT).

WHAT IT ENCODES

ABRUPT termination of the tachycardia with adenosine confirms an AV-node-DEPENDENT re-entry circuit — AVNRT or orthodromic AVRT. Adenosine breaks the loop by transiently blocking the node, which is both curative for the acute episode and diagnostic of the mechanism.

BOARD TRAP

After termination, the next step for prevention/definitive therapy depends on the substrate: catheter ablation is first-line definitive therapy for recurrent AVNRT/AVRT, and is strongly preferred in symptomatic WPW (where chronic AV-nodal-blocking drugs may be unsafe if AF develops).

11. AV block reveals atria

WHAT YOU SEE

An 'AV BLOCK' panel exposing sawtooth flutter waves and discrete P waves — the brief block doesn't stop the rhythm but unmasks the hidden atrial activity (flutter/focal AT).

WHAT IT ENCODES

When adenosine causes transient AV block WITHOUT terminating the rhythm, it UNMASKS the underlying atrial activity — exposing the sawtooth waves of atrial flutter (atrial rate ~300/min) or the discrete abnormal P waves of focal atrial tachycardia. This is the diagnostic use of adenosine in regular narrow-complex tachycardia.

BOARD TRAP

If adenosine reveals flutter/AT rather than terminating, the diagnosis (and therefore the treatment plan — rate control, anticoagulation considerations, possible ablation) changes; don't keep treating it as AVNRT.

12. Look for atrial activity

WHAT YOU SEE

A 'LOOK FOR ATRIAL ACTIVITY' banner over the panels — the instruction to watch the strip during the adenosine pause, when hidden flutter/AT waves briefly surface.

WHAT IT ENCODES

During the brief adenosine-induced AV pause, run a continuous rhythm strip and inspect for atrial activity — sawtooth flutter waves or atrial-tachy P waves that were hidden within the fast QRS complexes. Always record the strip BEFORE pushing adenosine so you capture the diagnostic moment.

BOARD TRAP

Failing to record/monitor during the pause wastes the diagnostic value of adenosine — the unmasking is transient (seconds), so capture it on a printed/continuous strip.

13. Unstable signs

WHAT YOU SEE

A cluster of red warning lights labelled hypotension, AMS, angina, HF — the four red alarms = the markers of hemodynamic instability that override the vagal/adenosine pathway.

WHAT IT ENCODES

The markers of hemodynamic instability in tachyarrhythmia are: HYPOTENSION/shock, ALTERED MENTAL STATUS, ischemic CHEST PAIN (angina), and acute HEART FAILURE (pulmonary edema). Any one of these in the context of the tachycardia defines an unstable patient.

BOARD TRAP

Instability must be attributable to the arrhythmia. Mild palpitations or anxiety alone are NOT instability — but true hypotension/AMS/ischemia/HF mandate immediate electrical therapy regardless of the rhythm subtype.

14. Unstable → cardiovert

WHAT YOU SEE

The red alarm-light cluster signalling emergency — instability with a pulse means skip the maneuvers and go straight to synchronized cardioversion.

WHAT IT ENCODES

Any UNSTABLE tachyarrhythmia with a pulse → immediate SYNCHRONIZED cardioversion. Sedate if time/consciousness allow. Synchronizing the shock to the R wave avoids delivering energy on the T wave (the vulnerable period), which could induce VF.

BOARD TRAP

Vagal maneuvers/adenosine first is WRONG when the patient is unstable. Also: use SYNCHRONIZED cardioversion (not unsynchronized defibrillation) for a perfusing rhythm — and pulseless VT/VF or polymorphic VT requires UNSYNCHRONIZED defibrillation instead.

15. Diving reflex / cold water

WHAT YOU SEE

A 'COLD WATER' face-immersion panel at the top of the VAGAL poster — splashing ice water on the face = the diving reflex, the preferred vagal maneuver in infants/young children.

WHAT IT ENCODES

Cold-water facial immersion (or an ice-water bag to the face) triggers the mammalian diving reflex, a powerful vagal stimulus that slows AV conduction. It is the preferred vagal maneuver in INFANTS and young children, in whom Valsalva and carotid massage are impractical.

BOARD TRAP

The diving reflex (ice to the face) is the go-to vagal maneuver in infants — carotid sinus massage is generally avoided in young children. Choosing CSM for a febrile infant with SVT is the wrong technique.

16. Gagging maneuver

WHAT YOU SEE

A 'GAGGING' figure at the bottom-left of the VAGAL poster — inducing a gag = another vagal trick that raises parasympathetic tone to slow AV conduction in stable SVT.

WHAT IT ENCODES

Inducing a gag reflex is another vagal maneuver that raises parasympathetic tone to slow AV-nodal conduction. Like the other maneuvers, it is reserved for STABLE, regular narrow-complex SVT and works only on AV-node-dependent rhythms.

BOARD TRAP

Like all vagal maneuvers, gagging will not convert atrial fibrillation, flutter, or VT, and is inappropriate as a first step in an unstable patient — who needs synchronized cardioversion.
